

Milestone 1: Literature Review and Exploratory Data Analysis

Group 9 Members

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Literature Review

Introduction

The rapid growth of e-commerce has transformed business–consumer interactions, supply-chain management and operations, while generating large volumes of transactional data. Decisions such as setting delivery promises (Cui et al., 2024) and predicting customer satisfaction (Zaghloul et al., 2024) have direct commercial consequences: delivery performance shapes customer satisfaction and purchase probability (Deshpande & Pendem, 2023; Ravula, 2023; Vakulenko et al., 2019), while customer satisfaction informs decisions that may improve profitability and reduce marketing costs (Zaghloul et al., 2024). These challenges have encouraged the adoption of machine learning (ML) for predictive analytics and data-driven decision-making in e-commerce (Dritsas & Trigka, 2025). This review examines ML approaches for delivery-performance and customer-satisfaction prediction, together with relevant modeling, feature-engineering and evaluation practices.

Existing Systems in Literature

Delivery performance is modeled through both classification (late or failed delivery) and regression (delivery lead time). Kandula et al. (2021) compared Random Forest (RF), XGBoost, LogitBoost, artificial neural networks and decision trees for delivery-success prediction, selecting XGBoost with a weighted loss for its lower prediction time and memory footprint, while achieving F1 scores of 0.72–0.80 with high precision but moderate recall.

For lead-time regression, Salari et al. (2022) extended quantile regression forests with an asymmetric loss on JD.com data to forecast delivery-time distributions and select cost-sensitive promises, while L. Zhang et al. (2023) developed a dual-graph multitask model on Taobao merchant, location and payment-time data to reduce underprediction of rare, long deliveries. Both highlight geographic and temporal predictors and long-tail evaluation. Beyond delivery prediction itself, delivery promises and actual delivery performance have also been linked to review ratings, satisfaction and subsequent purchasing behavior (Cui et al., 2024; Deshpande & Pendem, 2023; Ravula, 2023; Vakulenko et al., 2019).

Customer satisfaction has been modeled using both binary and ordinal approaches. Zaghloul et al. (2024) binarized review scores and compared Logistic Regression, Support Vector Machine (SVM), RF, Gradient Boosting Classifier (GBC), and Multilayer Perceptron (MLP) for predicting customer satisfaction. RF combined with random oversampling achieved the best performance, with an accuracy of 0.92, weighted F1-score of 0.92 and AUC-ROC of 0.90. Ordinal approaches, such as ordinal logistic regression, have also been applied to review scores (Akturk et al., 2022), since they preserve the ordinal structure of the rating scale that binarization discards. Deshpande and Pendem (2023) used a Heckman ordered probit model, which corrects for self-selection while treating the posted 1–5 rating as an ordered outcome.

Modeling Approaches

The train–test split should be made before any preprocessing, since fitting transformations on the full dataset leaks test information into training and overstates

real-world performance (Kapoor & Narayanan, 2023). Zaghloul et al. (2024) followed this with a random 67:33 split and standardized features using training-set statistics. For their delivery-prediction task, Kandula et al. (2021) instead held out the final week for testing and the preceding week for validation, mirroring how the model would be deployed.

Handling class imbalance is necessary because delivery outcomes are often skewed towards success. Kandula et al. (2021) compared oversampling (SMOTE, Borderline-SMOTE), undersampling (ENN, RENN), hybrids (SMOTE + ENN, SMOTE + Tomek) and a cost-sensitive weighted loss with weights inversely proportional to class frequency. Their final XGBoost model used the weighted loss, avoiding the need to generate or remove observations. Zaghloul et al. (2024), by contrast, reported best results with random oversampling. Van den Goorbergh et al. (2022) cautioned that for logistic regression, imbalance does not in itself impair probability estimation, and that resampling substantially worsens calibration without improving AUC-ROC. This favors cost-sensitive weighting and calibration-aware metrics over resampling.

Tree-based models are particularly relevant for structured tabular data. Grinsztajn et al. (2022) showed that tree-based models often outperform neural networks on typical tabular datasets, supporting their use for transactional e-commerce data. RF uses bagging and feature subsampling to reduce variance, while gradient-boosted models such as XGBoost, LightGBM and CatBoost build trees sequentially to improve predictive performance (X. Zhang et al., 2023). These models have also shown strong performance across different e-commerce applications, including delivery prediction (Kandula et al., 2021; Zaghloul et al., 2024) and sales and demand forecasting (Subashree & Savitha, 2026).

Feature Engineering Techniques

Feature engineering plays an important role in preparing e-commerce transactional data for modeling. In the reviewed studies, missing observations were often handled using complete-case deletion when the proportion of missing data was limited (Kandula et al., 2021; Zaghloul et al., 2024).

Domain-specific feature construction is widely used to transform raw transactional data into more meaningful predictors. Zaghloul et al. (2024) derived variables such as delivery-time differences, total order value and freight ratio, while Akturk et al. (2022) constructed operational measures including processing time, delivery time and early-or-late delivery relative to the promised date.

Feature selection can further reduce irrelevant or redundant predictors. Chandrashekar and Sahin (2014) distinguished filter, wrapper and embedded approaches. Filter methods rank features using statistical criteria such as correlation or mutual information; wrapper methods evaluate subsets based on model performance; and embedded methods perform selection during model training. Zaghloul et al. (2024), for example, used SelectKBest with mutual information to reduce 39 candidate features to 20.

Evaluation Metrics of ML Models

Classification models should be evaluated using multiple complementary metrics rather than accuracy alone, particularly when targets are imbalanced. Common metrics include AUC-ROC, AUC-PR, precision, recall, accuracy and F1-score (Kandula et al., 2021; Zaghloul et al., 2024). AUC-ROC evaluates the model's ability to discriminate between positive and negative classes across different classification thresholds, while AUC-PR places greater emphasis on the model's performance for the positive class. G-mean can further assess balance between sensitivity and specificity, while calibration should be assessed where predicted probabilities are to be interpreted as delivery risk (Van den Goorbergh et al., 2022).

For regression tasks, MAE and RMSE quantify prediction error, while R^2 measures the proportion of variance explained by the model. MAPE is also commonly used in forecasting studies such as Ensafi et al. (2022), although its suitability depends on the target scale.

Conclusion of Review

This review shows that tree-based ensembles, particularly RF and gradient-boosted models, have shown strong performance for delivery prediction and customer-satisfaction modeling because they capture nonlinear relationships in tabular data, especially when paired with domain-specific feature construction and selection. However, the studies differ in their use of random versus chronological splits and in imbalance treatment, give little attention to calibration and interpretability, and rarely report simple operational baselines, which makes results hard to compare and to relate to deployment. Phase 2 should therefore use time-aware validation, evaluate imbalance treatments carefully, include feature-importance or interpretability analysis, and compare against practical baselines such as promised delivery duration.

References

- Akturk, M. S., Mallipeddi, R. R., & Jia, X. (2022). Estimating impacts of logistics processes on online customer ratings: Consequences of providing technology-enabled order tracking data to customers. *Journal of Operations Management*, 68(6–7), 775–811. <https://doi.org/10.1002/joom.1204>
- Chandrashekar, G., & Sahin, F. (2014). A survey on feature selection methods. *Computers & Electrical Engineering*, 40(1), 16–28. <https://doi.org/10.1016/j.compeleceng.2013.11.024>
- Cui, R., Lu, Z., Sun, T., & Golden, J. M. (2024). Sooner or later? Promising delivery speed in online retail. *Manufacturing & Service Operations Management*, 26(1), 233–251. <https://doi.org/10.1287/msom.2021.0174>
- Deshpande, V., & Pendem, P. K. (2023). Logistics performance, ratings, and its impact on customer purchasing behavior and sales in e-commerce platforms. *Manufacturing & Service Operations Management*, 25(3), 827–845. <https://doi.org/10.1287/msom.2021.1045>
- Dritsas, E., & Trigka, M. (2025). Machine learning in e-commerce: Trends, applications, and future challenges. *IEEE Access*, 13, 99048–99067. <https://doi.org/10.1109/ACCESS.2025.3572865>
- Ensafi, Y., Amin, S. H., Zhang, G., & Shah, B. (2022). Time-series forecasting of seasonal items sales using machine learning—A comparative analysis. *International Journal of Information Management Data Insights*, 2(1), Article 100058. <https://doi.org/10.1016/j.ijime.2022.100058>
- Grinsztajn, L., Oyallon, E., & Varoquaux, G. (2022). Why do tree-based models still outperform deep learning on typical tabular data? *Advances in Neural Information Processing Systems*, 35, 507–520. <https://doi.org/10.52202/068431-0037>
- Kandula, S., Krishnamoorthy, S., & Roy, D. (2021). A prescriptive analytics framework for efficient e-commerce order delivery. *Decision Support Systems*, 147, 113584. <https://doi.org/10.1016/j.dss.2021.113584>
- Kapoor, S., & Narayanan, A. (2023). Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4, 100804. <https://doi.org/10.1016/j.patter.2023.100804>
- Ravula, P. (2023). Impact of delivery performance on online review ratings: The role of temporal distance of ratings. *Journal of Marketing Analytics*, 11(2), 149–159. <https://doi.org/10.1057/s41270-022-00168-5>

Salari, N., Liu, S., & Shen, Z.-J. M. (2022). Real-time delivery time forecasting and promising in online retailing: When will your package arrive? *Manufacturing & Service Operations Management*, 24(3), 1421–1436.

<https://doi.org/10.1287/msom.2022.1081>

Subashree, C., & Savitha, J. (2026). Enhancing predictive models in e-commerce: A comparative analysis using LightGBM, CatBoost and Prophet for sales and demand forecasting. *International Journal of Scientific Research in Computer Science, Engineering and Information Technology*, 12(1), 381–399.

<https://doi.org/10.32628/CSEIT2612139>

Vakulenko, Y., Shams, P., Hellström, D., & Hjort, K. (2019). Online retail experience and customer satisfaction: The mediating role of last mile delivery. *The International Review of Retail, Distribution and Consumer Research*, 29(3), 306–320.

<https://doi.org/10.1080/09593969.2019.1598466>

Van den Goorbergh, R., van Smeden, M., Timmerman, D., & Van Calster, B. (2022). The harm of class imbalance corrections for risk prediction models: Illustration and simulation using logistic regression. *Journal of the American Medical Informatics Association*, 29(9), 1525–1534. <https://doi.org/10.1093/jamia/ocac093>

Zaghloul, M., Barakat, S., & Rezk, A. (2024). Predicting e-commerce customer satisfaction: Traditional machine learning vs. deep learning approaches. *Journal of Retailing and Consumer Services*, 79, 103865.

<https://doi.org/10.1016/j.jretconser.2024.103865>

Zhang, L., Wang, M., Zhou, X., Wu, X., Cao, Y., Xu, Y., Cui, L., & Shen, Z. (2023). Dual graph multitask framework for imbalanced delivery time estimation. In *Database Systems for Advanced Applications (DASFAA 2023, Lecture Notes in Computer Science*, Vol. 13946, pp. 606–618). Springer.

https://doi.org/10.1007/978-3-031-30678-5_46

Zhang, X., Guo, F., Chen, T., Pan, L., Beliakov, G., & Wu, J. (2023). A brief survey of machine learning and deep learning techniques for e-commerce research. *Journal of Theoretical and Applied Electronic Commerce Research*, 18(4), 2188–2216.

<https://doi.org/10.3390/jtaer18040110>

Exploratory Data Analysis Report

Dataset Overview and Data Quality (Figure 1)

The dataset contains 99,441 orders across nine relational tables with 52 variables covering customer and seller location, order timestamps, product attributes, payments, and reviews. The tables form a star around orders: `order_items` links products and sellers, payments and reviews join on `order_id`, and customers and sellers reach geolocation through their zip-code prefix. `Order_items`, payments and reviews are one-to-many on `order_id` (1.14, 1.04 and 1.01 rows per order) and were aggregated before joining, so no join inflates the order count; all six business foreign keys match fully. A missing data assessment showed that review comment title (88.3% missing) and review comment message (58.7% missing) have the highest missing ratios and were excluded, retaining only the numeric review score. Category name is missing for 610 products (1.9%) and dimensions for 2 products, and delivery timestamps are missing for 2,965 orders (3.0%), mostly cancelled or in-transit orders. They were excluded from category-level and delivery-time analysis respectively. Duplicates were also resolved by aggregating geolocation to one coordinate per zip-code prefix, removing repeated review IDs, and summing payment rows to the order level. One further gap is structural rather than corrupt: 775 orders (0.8%) carry no line item, mostly unavailable or cancelled yet paid (R\$ 162,592) and rated 1.73 stars on average, so every item-level rate reported below assumes an order has at least one item.

Temporal Patterns (Figure 2)

Monthly orders grew ninefold, from 800 in January 2017 to a Black Friday peak of 7,544 in November 2017, then plateaued at 6,200–7,300 through 2018 (Figure 2a); the sparse edge months (349 orders) are set aside. A spike seen only once cannot be learned as seasonality, so Phase 2 should model it as an event flag. Orders also peak on Mondays, in the late morning and in the evening. Delivery is far less stable: 8.11% of 96,470 delivered orders arrived late, but the monthly rate ranges from 1.4% to 21.4% (Figure 2b). The median order is promised 23.2 days and delivered in 10.2, so punctuality rests on this cushion. The spikes of November 2017 and February–March 2018 came when delivery slowed to 13–14 days while promises did not lengthen, cutting the gap between the medians to 8–10 days (Figure 2c). Phase 2 therefore needs a time-ordered validation split.

Customer, Seller and Product Distributions (Figure 3)

Because `customer_id` is regenerated for every order, people are counted by `customer_unique_id`: 96,096 people placed the 99,441 orders and only 3.12% ordered more than once, which rules out churn and repeat-purchase problems. Both sides of the marketplace are concentrated, and supply more so than demand: São Paulo holds 42.0% of customer records but 60% of sellers, and the top three states hold 66.6% of customers and 79% of sellers, so most parcels leave the south-east regardless of where the buyer is. Selling is also concentrated within states: 130 of 3,095 sellers (4.2%) generate half of all sales value and the median seller has six orders. On the product side, the top ten of 73 categories cover 64.1% of listings and 63% of item revenue, led by health and beauty, watches and gifts, and bed, bath and table; two categories have no English translation. Long-tailed sellers and categories will therefore need grouping or smoothed encodings.

Correlation and Relationships (Figure 4)

Correlations are computed per order with Spearman's ρ , since monetary and physical variables are heavily skewed (Figure 4). Actual delivery time rises with seller–customer distance ($\rho = +0.54$), the promised lead time ($\rho = +0.52$) and freight cost ($\rho = +0.38$), and the late-delivery rate rises across both distance and freight quintiles (Figure 5a-b). Because Olist's own promise already captures much of the route signal, predicting the promise is the baseline any lead-time model must beat. Review score behaves differently: no purchase-time variable exceeds $\rho = -0.10$, while actual delivery time reaches $\rho = -0.23$, the strongest association it has with anything. That driver is observed only after purchase, making satisfaction a harder target than delivery. To prevent leakage, each of the 45 columns in the consolidated table is assigned one role: 6 identifiers, 26 known at purchase, 4 payment fields known shortly after, and 9 post-outcome columns, which hold the candidate targets and are never used as features.

Candidate Problem 1.1: Late-Delivery Classification (Figure 5)

Among delivered orders, 8.11% are late relative to the estimated delivery date, indicating an imbalanced classification target. Late-delivery risk is higher for cross-state orders and rises from 6.6% to 11.0% across seller–customer distance quintiles and from 6.1% to 9.3% across freight-cost quintiles. Order price and product weight show weaker positive associations, while freight ratio and item count show little systematic relationship; late rates also vary across product categories. These patterns identify temporal, geographic, logistics-cost and product-related variables as candidate features for classification.

Candidate Problem 1.2: Delivery Lead-time Regression (Figure 6)

Delivery lead time is measured as fractional days from purchase to customer receipt for 96,470 delivered orders with valid, non-negative durations. The distribution is right-skewed: median 10.22 days, mean 12.56 days and 90th percentile 23.10 days. All 306 orders above 60 days (0.32%) are retained (Figure 6a). Duration is positively associated with maximum observed seller–customer distance (Spearman $\rho = 0.541$; $n = 95,994$), promised duration ($\rho = 0.522$) and total freight ($\rho = 0.382$; Figure 6b). When sellers and customers are in the same state, median duration is 6.55 days versus 12.78 for cross-state orders. These unadjusted associations motivate candidate predictors without establishing causality or predictive accuracy. Using promised duration to predict actual duration provides a fixed, untrained baseline: MAE is 12.73 days and RMSE 15.12 days, with average overprediction of 11.18 days (Figure 6c). The promise therefore provides a useful signal but is not an accurate point estimate merely because it correlates with actual duration.

Inputs should be limited to order, location and quoted-promise information known at purchase; actual delivery, final status and reviews must be excluded. Phase 2 should compare models against the promise and training-only mean/median baselines on the same later test orders, with MAE as the primary criterion, RMSE and tail errors also reported, and log1p targets or robust losses assessed within training/validation.

Candidate Problem 2.1: Customer Satisfaction (Figure 7)

After aggregating item-level records to the order level and removing orders without a review score, 97,917 reviewed orders remained. Ratings were concentrated at the upper end, with 56,931 five-star reviews, while about 14.2% of orders scored 1–2.

Among the variables examined, delivery performance shows the clearest relationship with customer satisfaction. Mean review scores decline steadily as actual delivery duration increases, from 4.42 for orders delivered within six days to 3.42 for the longest-delivery group. The contrast between punctual and late deliveries is particularly large: on-time orders have an average rating of 4.29, compared with 2.57 for late orders. Product category and customer state also show variation in average ratings, suggesting that product- and location-related characteristics may provide additional predictive information. In comparison, freight cost appears to have a weaker relationship with satisfaction. Median freight values and freight-to-price ratios are similar across review scores, and the low-score rate varies only modestly and non-monotonically across freight-ratio groups.

Candidate Problem 2.2: Behavior by State (Figure 8)

Median distance between customer and seller explains most of the regional variation in price and financing, but not in service or payment habits. Freight-to-basket ratio ($\rho = +0.81$) and average order value ($\rho = +0.78$) rise almost monotonically with distance, from ~ 0.14 and $\sim R\$143$ in SP to ~ 0.28 and $\sim R\$220\text{--}265$ in the far North and Northeast. Installments follow more weakly ($\rho = +0.43$), with AP and RR as exceptions. Boleto share ($\rho = -0.01$) and late-delivery rate ($\rho = +0.01$) are unrelated to distance. RJ is as late as MA while remote RO, AP and AC are the most punctual. Review scores fall only modestly ($\rho = -0.30$), tracking late rate rather than distance. SP's weight pulls the national means (dashed) toward its profile, and the extreme values belong to low-volume states with noisy estimates.

Key Findings and Implications

1. Customer satisfaction is most strongly associated with delivery reliability among the operational factors examined. Late and longer deliveries receive substantially lower review scores, whereas freight-related measures show comparatively weak relationships with satisfaction. This makes delivery performance a priority feature for customer-satisfaction prediction.
2. Delivery promises contain useful information but do not accurately predict actual arrival times on their own. Although promised and actual delivery durations are positively associated, promised duration systematically overpredicts actual delivery time. Predictive models should therefore combine promise information with geography, order, logistics, and product characteristics.
3. Geographic and temporal conditions are major determinants of delivery performance. Seller–customer distance, cross-state fulfillment, freight cost, and purchase period are associated with delivery lead time and lateness. Their substantial temporal variation also supports time-ordered model validation.

Conclusion

The EDA supports three modelling directions: late-delivery classification, delivery lead-time regression, and customer-satisfaction prediction. Given the data-quality issues above, Phase 2 should use time-ordered validation, handle rare categories and skewed targets, and exclude post-outcome variables to prevent leakage.

GitHub link:

<https://github.com/IT5006-Group9/IT5006-Group9-Olist-Project>

Dashboard link:

<https://it5006-group9-olist-project-c5hmxth6jt543wszxqoaqd.streamlit.app/>

Appendix

Table	Rows	Rows per key	Missing values
orders	99,441	1 per order	approval 0.16%, carrier 1.79%, delivery 2.98%
order_items	112,650	1.14 per order	none
payments	103,886	1.04 per order	none
reviews	99,224	1.01 per order	comment title 88.3%, comment text 58.7%
customers	99,441	1 per order	none
sellers	3,095	1 per seller	none
products	32,951	1 per product	category 1.85%, dimensions 0.01%
geolocation	1,000,163	many per prefix	none (261,831 duplicate rows)
category_tr	71	1 per category	none

Figure 1. Dataset overview and data quality table

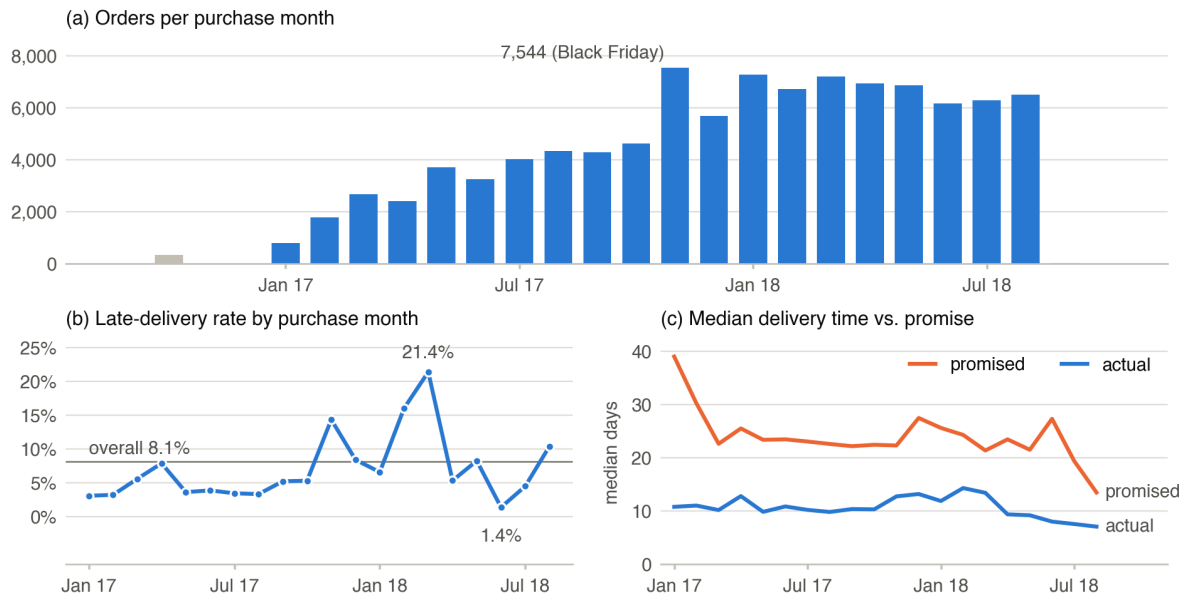


Figure 2. (a) Orders per purchase month; grey bars are sparse edge months. (b) Late-delivery rate and (c) Median actual versus promised delivery time, for months with at least 500 delivered orders.

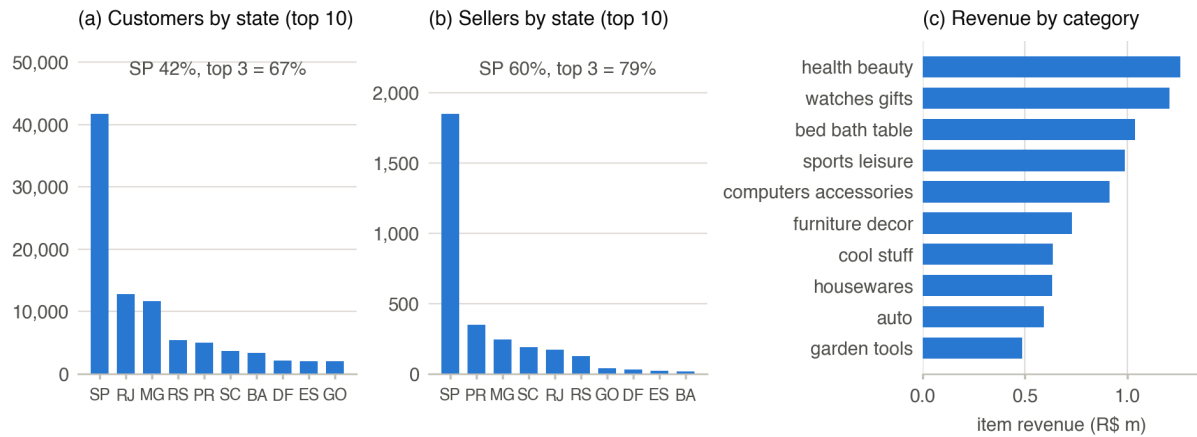


Figure 3. (a) Customers and (b) sellers by state, top 10 each. (c) Top 10 product categories by item revenue.

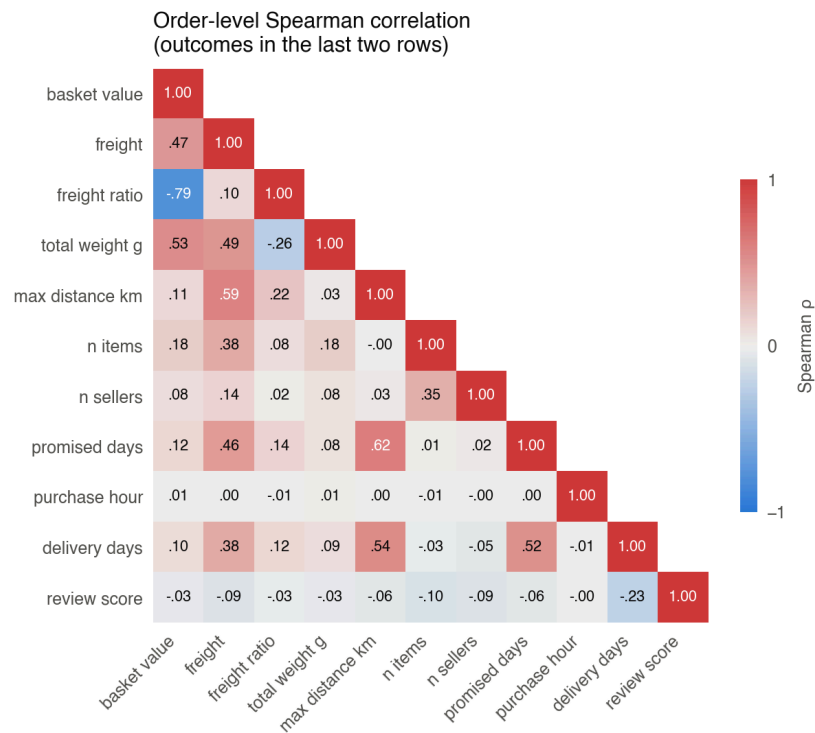


Figure 4. Order-level Spearman correlations between purchase-time variables and the two outcomes, lower triangle, 98,666 orders.

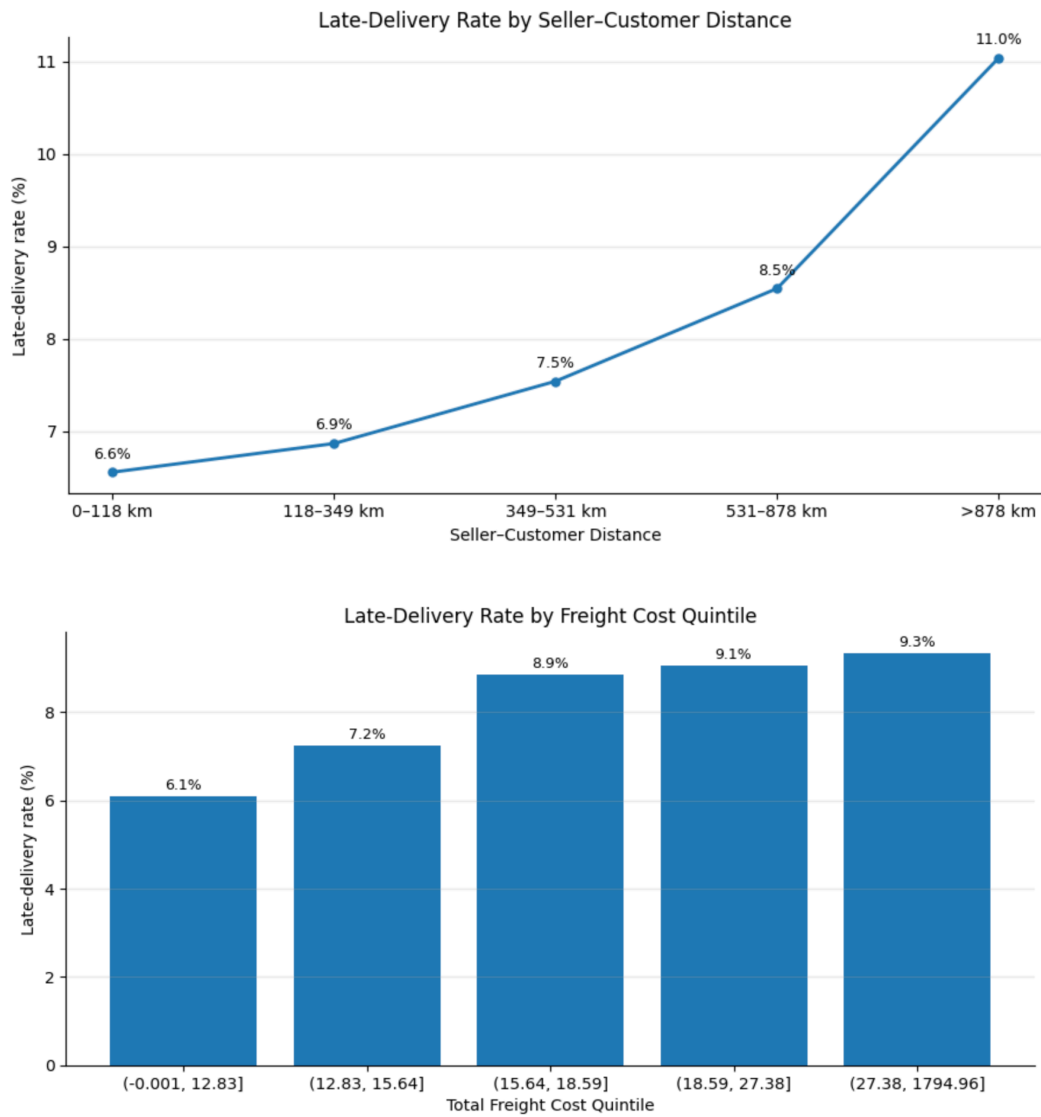


Figure 5. (a) Late-delivery rates by seller–customer distance quintile. (b) Late-delivery rates by freight-cost quintile.

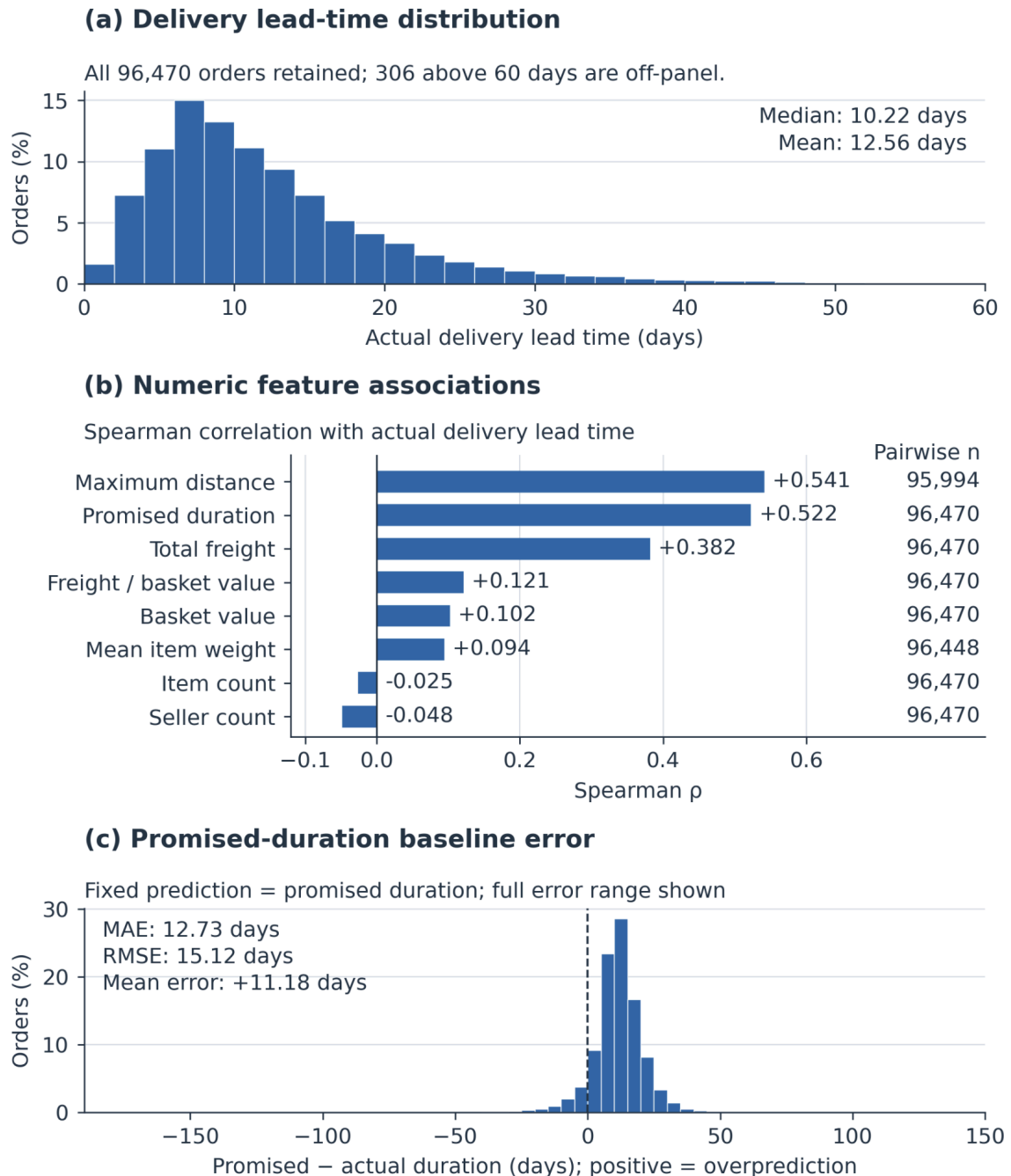
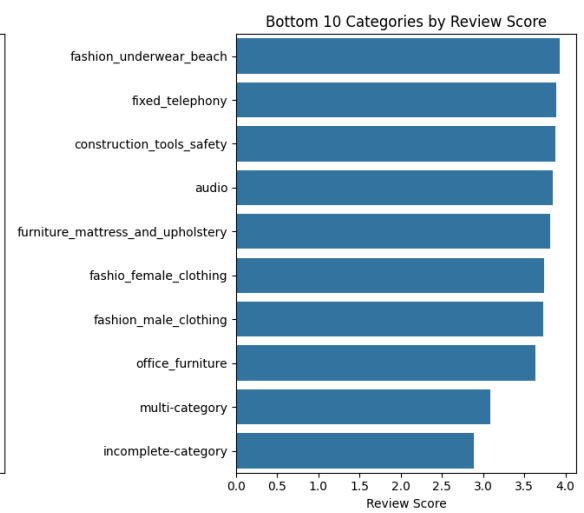
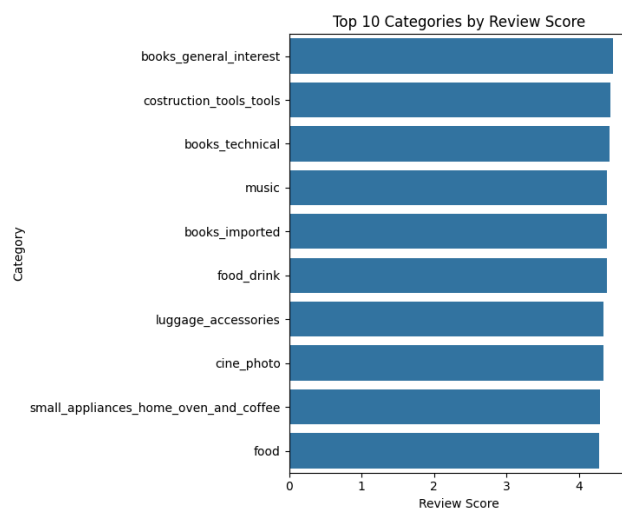
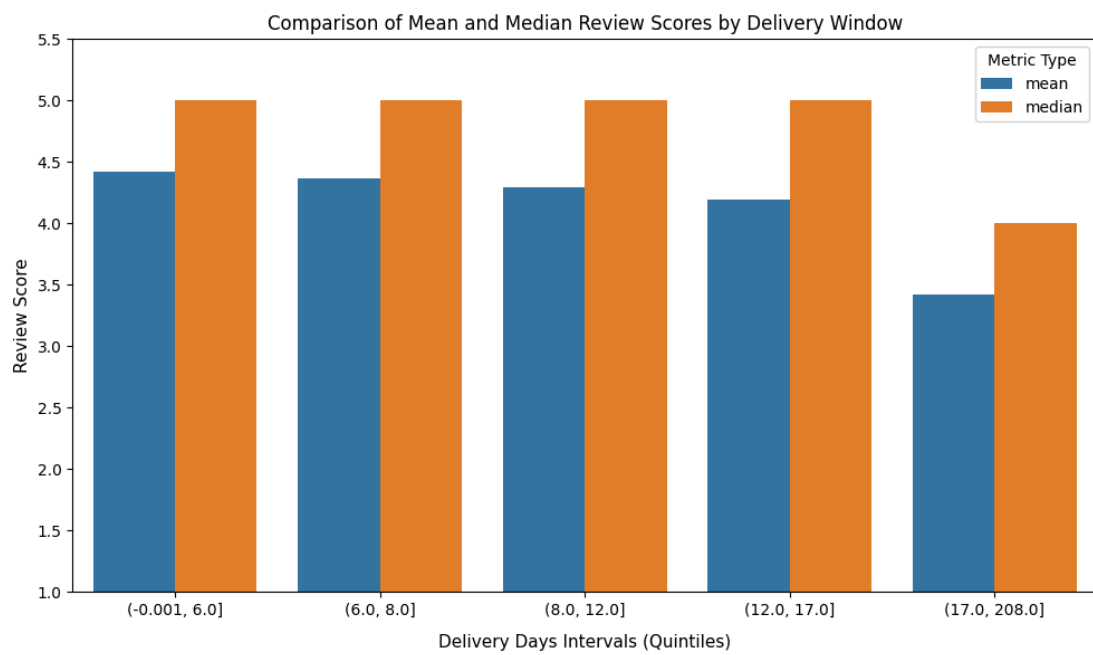
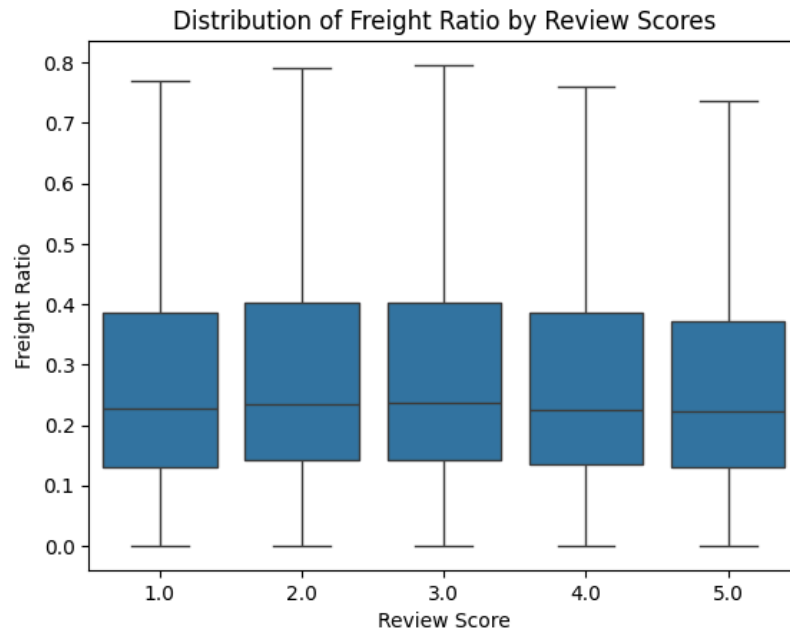


Figure 6. Delivery lead-time regression exploration for 96,470 eligible delivered orders: (a) duration distribution (0–60-day display; all 306 longer deliveries retained in analysis); (b) unadjusted Spearman correlations with pairwise sample sizes; (c) full-range errors from using promised duration to predict actual duration, defined as predicted minus actual days.





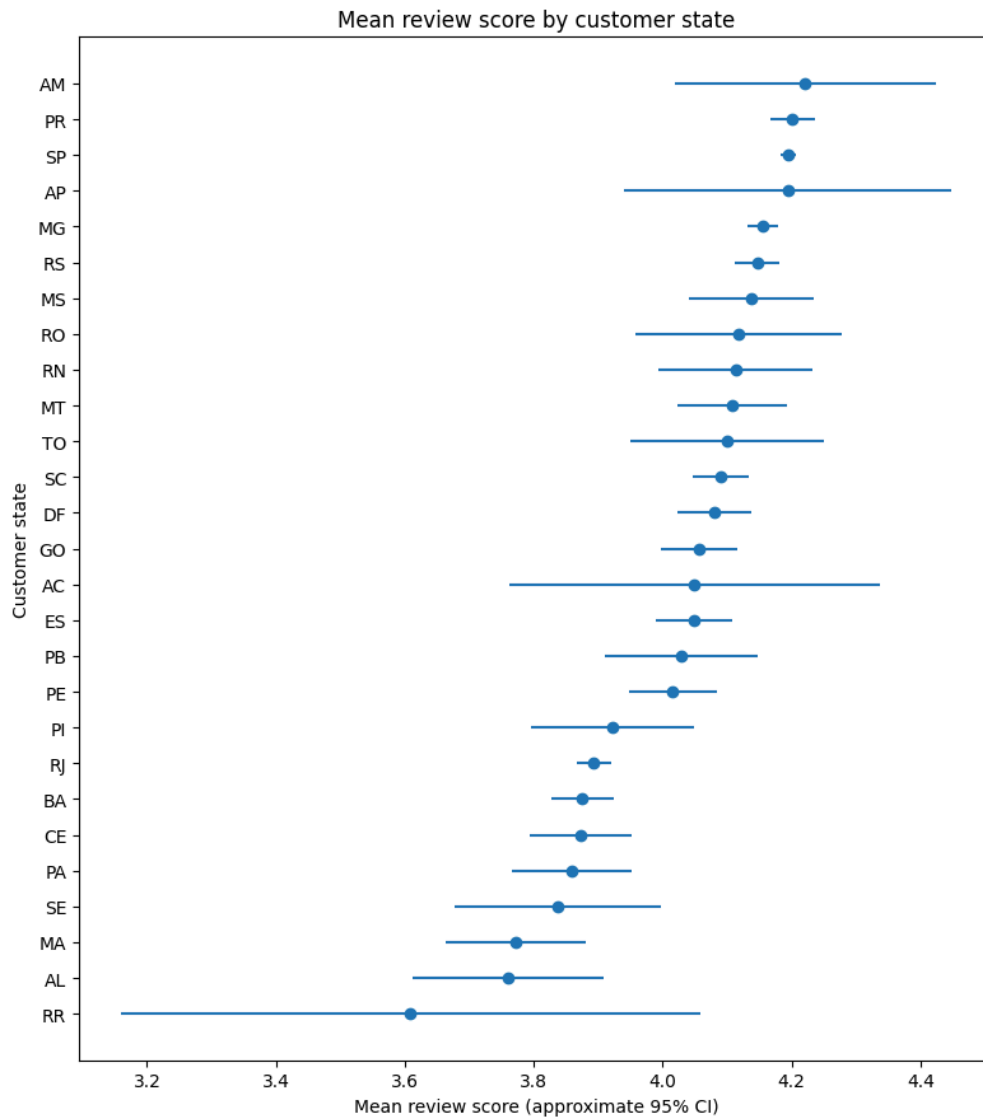


Figure 7. (a) Distribution of review scores. (b) Mean review score by each product category. (c) Distribution of freight ratio by review score. (d) Mean and median review score by delivery window. (e) Mean review score by customer state.

State-level behaviour vs. median shipping distance (point size = orders)

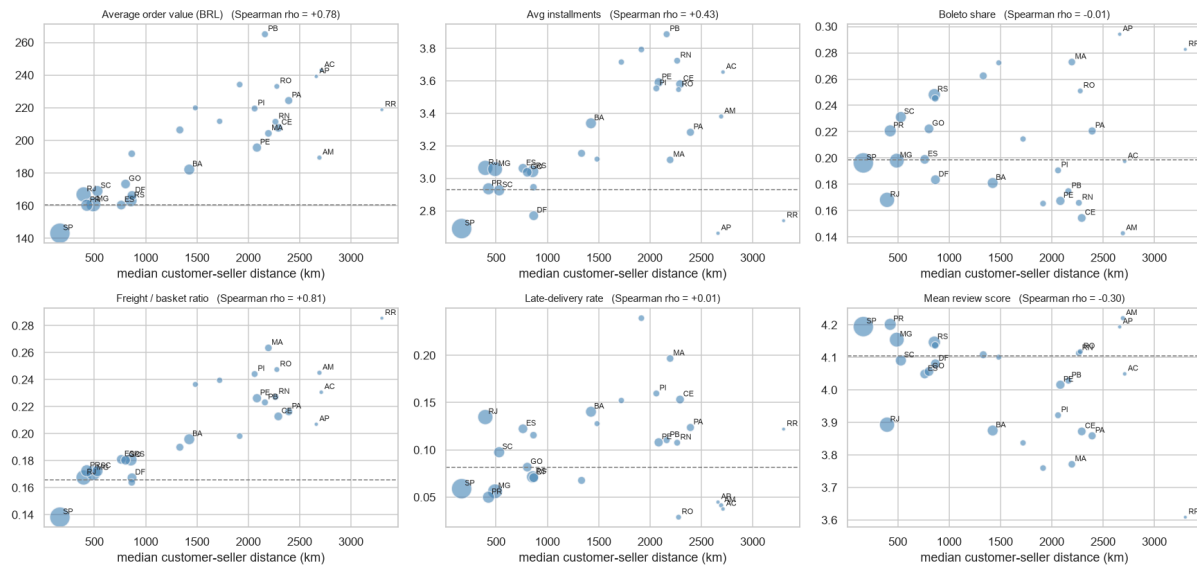


Figure 8. State-level behaviour against median customer–seller distance

AI Tool Declaration

We used GPT-5, Claude Fable 5.1 to generate and validate ideas, improve expression and refine our writing. We also used the models to debug our code. We are responsible for the content and quality of the submitted work.